

Performance Testing

Date	26 June 2026
Team ID	SWIID-2026-1243
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the responsiveness, stability, and scalability of **EduGenie – Google Gemini Powered Learning Assistant** under normal and peak workloads. The testing focused on AI-powered educational services, including intelligent question answering, concept explanation, quiz generation, educational text summarization, and personalized learning path recommendations. The objective was to ensure that the FastAPI backend and Google Gemini integration deliver accurate responses with minimal latency while maintaining optimal system resource utilization.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing & API Performance Testing
Target Module / API	Question Answering API, Quiz Generation API, Summarization API, Learning Recommendation API
Test Environment	Local Machine (FastAPI + Python 3.11 + Google Gemini API)
Test Date	26 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1		10	60	Accurate response within 2 seconds

	Generate AI response for educational questions			
2	Generate concept explanations using Gemini	15	60	Consistent explanations without failures
3	Generate quizzes for selected topics	20	60	Quiz generated successfully for all requests
4	Summarize educational content and recommend learning paths	20	60	Stable performance with accurate AI responses



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 sec	Pass	Meets performance target
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Stable during peak requests
3	Throughput (Req/sec)	> 20 Req/sec	27 Req/sec	Pass	Efficient request handling
4	Error Rate	< 1%		Pass	

			0%		No failed API requests
5	CPU Utilization	< 80%	56%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	61%	Pass	Stable memory consumption

Step 4: Observations & Analysis

Key Findings:
 [Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
 [List any performance bottlenecks observed during testing]

Optimization Steps Taken:
 [Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]